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Circuit board device for inserting an expansion card

Abstract

A circuit board device is adapted for inserting an expansion card with an electrical end and a locking part. The circuit board device includes a circuit board body, a slot, and a linkage structure located beside the slot. The linkage structure includes a connecting rod with a first end, a second, and an axis, a pressing part, and a fixing part. The pressing part is connected to the first end and deviates from the axis. The fixing part is located at the second end and has a hook, an elastic arm close to the slot and the hook, and a fixing member. The hook deviates from the axis and has a first slope facing away from the circuit board body. The fixing member extends from the spring arm towards the slot and includes a second slope facing away from the circuit board body and a third slope facing the hook.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application serial no. 112204459, filed on May 8, 2023. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to a circuit board device, and in particular to a circuit board adapted for

inserting an expansion card.

Description of Related Art

(3) With the advancement of technology, the functions of electronic devices are becoming more and more diverse, and there is a need for more scalability. The direction of this research focuses on the way to quickly and conveniently configure the expansion card or remove the expansion card from the circuit board.

SUMMARY

(4) A circuit board device of the disclosure is adapted for inserting an expansion card. The expansion card includes an electrical end and a locking part. The circuit board device includes a circuit board body, a slot, and a linkage structure. The slot is disposed on the circuit board body. The linkage structure is rotatably disposed on the circuit board body and is located beside the slot. The linkage structure includes a connecting rod, a pressing part, and a fixing part. The connecting rod has a first end and a second end opposite to each other and an axis. The pressing part is connected to the first end and deviates from the axis. The fixing part is located at the second end and has a hook, an elastic arm, and a fixing member. The hook is close to the slot and deviates from the axis, and has a first slope facing away from the circuit board body. The elastic arm is close to the slot and the hook. The fixing member extends from the elastic arm towards the slot and includes a second slope facing away from the circuit board body and a third slope facing the hook. When the electrical end is inserted in the slot, the locking part pushes against the first slope, so that the linkage structure rotates around the axis, and the hook gives way, and the locking part pushes against the second slope, and the elastic arm deforms, so that the fixing member gives way, and the expansion card is fixed between the fixing member and the hook.

(5) As stated above, the linkage structure of the circuit board device of the disclosure is rotatably disposed on the circuit board body and is located beside the slot. The pressing part of the linkage structure is connected to the first end of the connecting rod and deviates from the axis. The hook is close to the slot and deviates from the axis, and has the first slope facing away from the circuit board body. The fixing member extends from the elastic arm towards the slot and includes the second slope facing away from the circuit board body and the third slope facing the hook. When the electrical end of the expansion card is inserted in the slot, the locking part of the expansion card pushes against the first slope, so that the linkage structure rotates around the axis, and the hook gives way, and the locking part pushes against the second slope, and the elastic arm deforms, so that the fixing member gives way, and the expansion card is fixed between the fixing member and the hook, which is quite convenient to fix the expansion card.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic view of a circuit board device according to an embodiment of the disclosure.

(2) FIG. 2 is an enlarged schematic view of a linkage structure in FIG. 1.

(3) FIG. 3 is an exploded view of FIG. 2

(4) FIG. 4 is a partial enlarged schematic view of a fixing part in FIG. 3.

(5) FIG. 5 is a partial enlarged schematic view of FIG. 1.

(6) FIG. 6 is a schematic view of the linkage structure and a locking part in FIG. 5 from another viewing angle.

(7) FIG. 7 and FIG. 8 are partial enlarged schematic views of the locking part moving down and the linkage structure.

(8) FIG. 9 is a schematic view of the linkage structure and the locking part in FIG. 8 from another viewing angle.

(9) FIG. 10 is a partial enlarged schematic view of the locking part and the fixing part after a pressing part is pressed.

(10) FIG. 11 is a schematic view of a fixing part in FIG. 1 according to another embodiment of the disclosure.

DESCRIPTION OF THE EMBODIMENTS

(11) Referring to FIG. 1, a circuit board device 20 in this embodiment includes a circuit board body 22, a slot 24, and a linkage structure 100. The slot 24 is disposed on the circuit board body 22. The circuit board body 22, for example, is a mainboard, and the slot 24, for example, is a Peripheral Component interconnect Express (PCI-E) slot, but the types of the circuit board body 22 and the slot 24 are not limited.

(12) The slot 24 is adapted for inserting an expansion card 10. In this embodiment, the expansion card 10 takes a graphics card as an example, but the type of the expansion card 10 is not limited. The expansion card 10 includes an electrical end 12 and a locking part 14. The electrical end 12 is inserted in the slot 24, and the locking part 14 is used for confining the expansion card 10 to the slot 24, so that the expansion card 10 is not easily detached from the slot 24.

(13) The linkage structure 100 is rotatably disposed on the circuit board body 22 and is located beside the slot 24. In FIG. 2, the linkage structure 100 includes a connecting rod 110, a pressing part 120, and a fixing part 130. The connecting rod 110 has a first end 112 and a second end 114 opposite to each other and an axis 116. The pressing part 120 is connected to the first end 112 and deviates from the axis 116. Specifically, there is a connecting segment 170 between the connecting rod 110 and the pressing part 120. The pressing part 120 is connected to the first end 112 through the connecting segment 170. As the connecting rod 110, the connecting segment 170, and the pressing part 120 are respectively bent to be connected, the pressing part 120 deviates from the axis 116.

(14) Referring to FIG. 1 and FIG. 2, in this embodiment, the circuit board device 20 also includes a power slot 30 disposed on the circuit board body 22. The first end 112 (FIG. 2), the connecting segment 170 (FIG. 2), and the pressing part 120 (FIG. 2) form a space S. The power slot 30 is located in the space S. That is, in this embodiment, the pressing part 120 deviates from the axis 116 to give way to the power slot 30.

(15) Furthermore, in this embodiment, the expansion card 10 also includes a power terminal 16 aligned with the power slot 30. When the electrical end 12 of the expansion card 10 is inserted in the slot 24, the power terminal 16 is inserted in the power slot 30 at the same time, so that the circuit board body 22 transmits signals and supplies power to the expansion card 10. Of course, in other embodiments, the space S may also be configured with other components, and there is no limitation.

(16) Referring to FIG. 3, in this embodiment, circuit board device 20 also includes a stand 180 and a torsional spring 185. The stand 180 is fixed to the circuit board body 22 (FIG. 1). The connecting rod 110 is rotatably disposed on the stand 180 with the axis 116 as the axis of rotation. The torsional spring 185 is located between the stand 180 and the connecting rod 110, two ends of the torsional spring 185 are connected to the stand 180, and a central part of the torsional spring 185 is connected to the connecting rod 110, so that the linkage structure 100 may be taken back by the torsional spring 185 after being rotated relative to the stand 180. In other embodiments, one end of the torsional spring 185 may also be connected to the stand 180, and the other end may be connected to the connecting rod 110.

(17) Additionally, in this embodiment, the material of the linkage structure 100 is plastic, for example. To enhance the structural strength, the material of the connecting segment 170 or the pressing part 120 may be metal.

(18) Referring to FIG. 4, the fixing part 130 of the linkage structure 100 is located at the second end 114 and has a hook 132, an elastic arm 140, and a fixing member 150. The hook 132 is close to the slot 24 (FIG. 1) and deviates from the axis 116, and has a first slope 138 facing away from the

circuit board body **22** (FIG. 1). The elastic arm **140** is close to the slot **24** (FIG. 1) and the hook **132**. The elastic arm **140** has a fixing end close to the hook **132** and a free end away from the hook **132**.

(19) The fixing member **150** extends from the elastic arm **140** towards the slot **24** (FIG. 1) and includes a second slope **152** facing away from the circuit board body **22**, a third slope **154** facing the hook **132**, and an abutting surface **156** opposite to the third slope **154**. The hook **132** includes a handle part **134** and a hook part **136**. An end of the elastic arm **140** is connected to the handle part **134**. The third slope **154** faces the handle part **134**. The hook part **136** extends from the handle part **134** towards the fixing member **150**.

(20) Referring to FIG. 5 and FIG. 6, at this time, the locking part **14** of the expansion **10** is still above the third slope **154** of the hook **132** and is not yet in contact with the third slope **154**. As shown in FIG. 6, in this embodiment, the hook **132** is located on the top left of the axis **116**, not directly above the axis **116** to be staggered from the axis **116**. This design enables the linkage structure **100** to rotate with the axis **116** as the axis of rotation when the locking part **14** subsequently pushes down the first slope **138** of the hook **132**. Additionally, as presented in FIG. 5, the slot **24** has an extension line **25**. The extension line **25** is staggered from and parallel to the axis **116**. That is, the position of the expansion card **10** after being inserted in the slot **24** is staggered from the axis **116**, so that the hook **132** rotated back can limit the position of the locking part **14**. The following is a detailed description of the locking process.

(21) As shown in FIG. 7, the locking part **14** moves down and pushes against the first slope **138**, so that the linkage structure **100** rotates with the axis **116** as the axis of rotation, and the hook **132** gives way. As the linkage structure **100** rotates, the fixing member **150** rotates below the locking part **14**. When moving down, the locking part **14** pushes against the second slope **152** of the fixing member **150**. Since the fixing member **150** is located on the elastic arm **140**, the elastic arm **140** deforms backwards, so that the fixing member **150** gives way, and the locking part **14** can continue moving down. In this process, the torsional spring **185** (FIG. 3) accumulates elasticity, and the expansion card **10** continues moving down until the electrical end **120** is inserted in the slot **24**.

(22) As presented in FIG. 8 and FIG. 9, when the locking part **14** of the expansion card **10** moves down without being in contact with the first slope **138** of the hook part **136** and the second slope **152** of the fixing member **150**, the linkage structure **100** is not interfered by the locking part **14**. The torsional spring **185** (FIG. 3) releases the accumulated elasticity and brings the linkage structure **100** back to the original position, so that the hook **132** reversely rotates back to the top of the locking part **14**. At this time, the hook **132** limits the locking part **14** from moving towards a first direction D1 (FIG. 8) away from the circuit board body **22**, and the locking part **14** is located beside the third slope **154** and is limited by the fixing member **150** in a second direction D2 (FIG. 8), so that the expansion card **10** is fixed between the fixing member **150** and the hook **132**.

(23) As shown in FIG. 9, at this stage, the locking part **14** of the expansion card **10** is located between the handle part **134** and the third slope **154**. The hook **132** and the locking part **14** deviate from the axis **116** in the up and down direction and are located at the same side of the axis **116**. At this time, if the hook part **136** unexpectedly moves up, the hook part **136** generates a torque about the axis **116** in a clockwise direction, the locking part **14** pushes the hook part **136** upwards, and the hook part **136** also generates a torque about the axis **116** in the clockwise direction, so that the hook **36** can be hooked tightly without being pushed upwards by the locking part **14** to move away from the top of the locking part **14**. Therefore, the expansion card **10** (FIG. 1) can be fixed securely.

(24) To lift the limitation on the position of the locking part **14** by the linkage structure **100**, only the pressing part **120** of FIG. 1 needs to be pressed. The pressing part **120** is forced to rotate the linkage structure **100** with the connecting rod **110** as the axis, the hook **132** gives way and lifts the limitation on the position of the locking part **14** in the first direction D1, and the third slope **154** (FIG. 9) moves along the locking part **14**, so that the elastic arm **140** deforms and gives way, and the fixing member **150** lifts the limitation on the position of the locking part **14** in the second

direction D2 (FIG. 8).

(25) After the third slope **154** leaves the locking part **14** and the elastic arm **140** returns, as shown in FIG. **10**, the abutting surface **156** of the fixing part **130** abuts against the locking part **14**, so that the linkage structure **100** (FIG. **10**) maintains a rotary state. This design prevents a user from pressing the pressing part **120** all the time, and the connecting rod **110** maintains the rotary state, so that the user may slowly pull out the expansion card **10**.

(26) Referring to FIG. **11**, the difference between a fixing part **130a** in this embodiment and the fixing part **130** of the foregoing embodiment is that the fixing part **130a** in this embodiment also includes a limiting part **160** located on the elastic arm **140**, the fixing member **150** is located between the hook part **136** and the limiting part **160**, and the fixing member **150** is closer to the circuit board body **22** (FIG. **1**) than the hook part **136** and the limiting part **160**. The limiting part **160** is used for limiting a track of the locking part **14** when moving down to prevent the locking part **14** from being too far away from the hook part **136** and unable to be pushed down to the hook part **136** to rotate the hook part **136**.

(27) As described above, the linkage structure of the circuit board device in the disclosure is rotatably disposed on the circuit board body and is located beside the slot. The pressing part of the linkage structure is connected to the first end of the connecting rod and deviates from the axis. The hook is close to the slot and deviates from the axis, and has the first slope facing away from the circuit board body. The fixing member extends from the elastic arm towards the slot and includes the second slope facing away from the circuit board body and the third slope facing the hook. When the electrical end of the expansion card is inserted in the slot, the locking part of the expansion card pushes against the first slope, so that the linkage structure rotates around the axis, and the hook gives way, and the locking part pushes against the second slope, and the elastic arm deforms, so that the fixing member gives way. When the linkage structure returns, the hook restricts the locking part from moving in the first direction away from the circuit board body, and the locking part is located beside the third slope and is limited in the second direction, so that the expansion card is fixed between the fixing member and the hook. Therefore, the expansion card can be fixed conveniently and securely.

(28) In addition, when the pressing part is forced, so that the linkage structure rotates with the connecting rod as the axis, the hook gives way and lifts the limitation on the position of the locking part in the first direction, and the third slope moves along the locking part, so that the elastic arm deforms, and the fixing member lifts the limitation on the position of the locking part in the second direction. As the pressing part is away from the slot, the pressing part is not blocked by structures near the slot, so the user can unlock by pressing smoothly. Therefore, the expansion card can be easily removed. Additionally, as the pressing part deviates from the axis, when the pressing part is pressed downwards, the linkage structure lifts the limitation on the position of the locking part by rotating so the whole structure is relatively simple and requires fewer components.

Claims

1. A circuit board device, adapted for inserting an expansion card, wherein the expansion card comprises an electrical end and a locking part, the circuit board device comprising: a circuit board body; a slot, disposed on the circuit board body; and a linkage structure, rotatably disposed on the circuit board body and located beside the slot, the linkage structure comprising: a connecting rod, having a first end and a second end opposite to each other and an axis; a pressing part, connected to the first end and deviating from the axis; and a fixing part, located at the second end and having: a hook, close to the slot and deviating from the axis, and having a first slope facing away from the circuit board body; an elastic arm, close to the slot and the hook; and a fixing member, extending from the elastic arm towards the slot and comprising a second slope facing away from the circuit board body and a third slope facing the hook, wherein when the electrical end is inserted in the slot,

the locking part pushes against the first slope, so that the linkage structure rotates around the axis, and the hook gives way, and the locking part pushes against the second slope, and the elastic arm deforms, so that the fixing member gives way, and the expansion card is fixed between the fixing member and the hook.

2. The circuit board device according to claim 1, wherein when the linkage structure returns, the hook restricts the locking part from moving in a first direction away from the circuit board body, and the locking part is located beside the third slope and is limited in a second direction by the fixing member, so that the expansion card is fixed between the fixing member and the hook.

3. The circuit board device according to claim 2, wherein when the pressing part is forced to rotate the linkage structure with the connecting rod as an axis, the hook gives way and lifts limitation on a position of the locking part in the first direction, and the third slope moves along the locking part, so that the elastic arm deforms and gives way, and the fixing member lifts limitation on a position of the locking part in the second direction.

4. The circuit board device according to claim 1, wherein the fixing member further comprises an abutting surface opposite to the third slope, and after the third slope leaves the locking part and the elastic arm returns, the abutting surface abuts against the locking part, so that the linkage structure maintains a rotary state.

5. The circuit board device according to claim 1, wherein the hook comprises a handle part and a hook part, an end of the elastic arm is connected to the handle part, the third slope faces the handle part, and the hook part extends from the handle part to the fixing member.

6. The circuit board device according to claim 5, wherein when the electrical end of the expansion card is inserted in the slot and the linkage structure returns, the locking part of the expansion card is located between the handle part and the third slope, and the hook and the locking part are located at a same side of the axis.

7. The circuit board device according to claim 5, wherein the fixing part further comprises a limiting part located on the elastic arm, the fixing part is located between the hook part and the limiting part, and the fixing member is closer to the circuit board body than the hook part and the limiting part.

8. The circuit board device according to claim 1, wherein the slot comprises an extension line, and the extension line is staggered from and parallel to the axis.

9. The circuit board device according to claim 1, further comprising a power slot disposed on the circuit board body, there is a connecting segment between the connecting rod and the pressing part, the first end, the connecting segment, and the pressing part form a space, and the power slot is located in the space.

10. The circuit board device according to claim 1, further comprising a stand fixed to the circuit board body, and the connecting rod is disposed on the stand.

11. The circuit board device according to claim 10, wherein the stand is provided with a torsional spring, and the torsional spring is connected between the stand and the connecting rod.
